

Muhammad Saad Shabir

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EDUCATION

Carleton University

Bachelor of Information Technology - Network Technology

Ottawa, ON

Expected Apr 2028

SKILLS

Languages: C++, Go, Rust, Python, C, Bash, SQL

Systems & Tools: Linux, Docker, Kubernetes, AWS, Git, CMake, Containerlab, Prometheus

Networking & Security: TCP/IP, Cisco IOS, VLANs, OSPF, ECMP, ACLs, eBPF, FRRouting, Wireshark

EXPERIENCE

AriesTECH

Network Technician

Ottawa, ON

Jan 2024 – Apr 2025

- Configured 10 Cisco routers and switches at one site with VLANs, trunking, inter-VLAN routing, ACLs, static routes, RIP, and OSPF.
- Maintained 99.9% uptime through monitoring, Wireshark/Cisco IOS troubleshooting, firmware updates, ACL reviews, and backups.
- Wrote Python scripts for configuration and documentation tasks, cutting manual setup time by 30% and reducing deployment errors.

SELECTED PROJECTS

ZTAP – Network Policy Enforcement ↗

Go, eBPF, Kubernetes, Linux

- Built a Go Kubernetes node agent that compiles NetworkPolicy into per-container eBPF rules for IPv4 TCP/UDP traffic.
- Resolved pod and namespace selectors into ingress/egress rules, with live eBPF flow events and Prometheus metrics.
- Measured 180 ms p95 reconciliation on a 250-Pod, 2,500-rule Linux test fixture with cached inputs and real eBPF enforcement.

RouteProof – Routing Simulation and Reachability Verification ↗

C++20, OSPF, ECMP, FRRouting, Containerlab, Python

- Built a C++20 OSPF-style simulator with ECMP routing and deterministic replay of link and router failures.
- Checked reachability across every ECMP branch and compared route costs and next hops against FRRouting labs.
- Measured scenario runtime, memory, and scaling with repeatable benchmarks and published raw results.

NetScope – Packet and Flow Analysis ↗

Rust, libpcap, BPF, PCAP, Chart.js

- Wrote a Rust CLI for live capture and offline PCAP analysis, with TCP flow tracking, RTT estimates, and packet exports.
- Added SYN flood and port scan checks, BPF filters, and a local dashboard with live traffic views and Prometheus metrics.
- Benchmarked 1M-packet offline PCAPs at 3.5M packets/s median on Apple M4 with four workers and zero dispatch drops.